Chapter 15: Probability and Inference in Practice

The Practice of Statistics in the Life Sciences
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Part III: Foundations of Inference

Probability and Inference Belong Together

Chapters 9–12 developed probability models for uncertain outcomes. Chapters 13–14 used those models to describe sampling variation and draw conclusions about populations.

Chapter 15 brings these ideas together. The same probability rules and distributions now help us decide whether an inferential conclusion is reliable and useful.

Central Connection

Probability describes what could happen under a model. Inference compares the observed data with those model-based possibilities.

Chapter Roadmap

The chapter follows the path from a probability model to a practical conclusion.

1. Connect probability rules and distributions to inferential reasoning
2. Assess the study design and the assumptions behind the model
3. Examine how confidence level, variability and sample size affect estimation
4. Interpret hypothesis tests, error probabilities and practical importance
5. Plan studies using a target margin of error or power

The examples include continuous measurements, binary outcomes and event counts.

Parameters, Statistics and Randomness

A population parameter is a fixed but usually unknown number. A sample statistic is calculated from the observations and changes from sample to sample.

Question	Parameter	Statistic
Population mean	μ	$\bar{x}$
Population probability	p	$\hat{p}$
Population event rate	λ	Observed rate or count

Inference uses the observed statistic to learn about the parameter. Probability describes how much the statistic could vary before the sample is selected.

Sampling Distributions Supply Inferential Probabilities

A probability model for individual outcomes does not directly describe the behaviour of a sample statistic.

The sampling distribution connects the two levels:

1. A population model describes possible individual outcomes
2. Repeated random samples produce different statistics
3. The sampling distribution describes those possible statistic values
4. Confidence intervals and tests use that distribution to measure uncertainty

Key Distinction

The distribution of individual observations and the sampling distribution of a statistic answer different questions.

The Probability-to-Inference Process

Stage	Question	Statistical object
Probability model	What outcomes could occur if the model were correct?	Normal, binomial or Poisson distribution
Sampling distribution	How would a statistic vary across repeated samples?	Distribution of $\bar{x}$, $\hat{p}$ or a count
Inference	Is the observed statistic plausible under a stated population claim?	Confidence interval or P-value
Practical conclusion	What does the evidence support in the study setting?	Contextual conclusion with limitations

Inference remains conditional on the probability model and the data-collection process.

Probability Rules Inside Inference

Several rules from Chapters 9–10 appear directly in inferential calculations.

- **Complement rule:** upper-tail probabilities use $P(Z \geq z) = 1 - P(Z < z)$
- **Addition rule:** a two-sided P-value combines probabilities from two disjoint tails
- **Conditional probability:** a P-value is calculated conditional on H_0 and the model assumptions being true
- **Independence:** standard formulas usually assume that one observation does not determine another

Important Distinction

$P(\text{data as or more extreme} \mid H_0)$ is not the same as $P(H_0 \mid \text{data})$.

The Normal Distribution in Inference

Chapter 11 used z-scores to convert observations to areas under the Standard Normal curve. In inference, the same process applies to a statistic.

$$z = \frac{\text{statistic} - \text{model value}}{\text{standard error}}$$

- Critical values identify central areas used for confidence intervals
- Tail areas measure how unusual a test statistic is under H_0
- The standard error replaces the standard deviation of individual observations

The Normal model therefore connects standardization, confidence intervals and P-values.

Binomial and Poisson Models in Inference

The distribution should match the type of outcome being studied.

Model	**Data structure**	**Inferential question**
Binomial	Number of successes in n independent trials with the same success probability	Is the population success probability consistent with a proposed value?
Poisson	Number of events in a fixed amount of time, area or exposure under a stable rate	Is the event rate consistent with a proposed value?

The exact distribution can provide a tail probability. A Normal approximation may be used when its conditions are satisfied.

Exact Models and Normal Approximations

An inferential probability can come from the original probability model or from an approximation to a sampling distribution.

Approach	**Advantage**	**Important condition**
Exact binomial or Poisson calculation	Retains the discrete probability model	The model assumptions must fit the data-generating process
Normal approximation	Uses standardization and Normal tail areas	The sample or expected counts must be sufficiently large

The word exact refers to the calculation under the model. It does not guarantee that the model itself is correct.

Conditions for One-Sample z Procedures

The one-sample z procedures for a population mean assume that

- The observations come from a random sample or a defensible random process
- The observations are independent
- The population is Normal, or the sampling distribution of $\bar{x}$ is approximately Normal
- The population standard deviation σ is known

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}} \qquad \text{and} \qquad z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

Later chapters replace the uncommon known-σ assumption with an estimate from the sample.

The Source of the Data

Probability can describe sampling variation only when the data arise from a process that the probability model represents reasonably well.

Chance can enter a study through

- **Random sampling**, which supports generalization from the sample to a defined population
- **Random assignment**, which helps balance lurking variables and supports causal comparisons

First Question

Before calculating an interval or a P-value, identify how the observations entered the study and which conclusion that design can support.

Random Sampling and Random Assignment

Design feature	Main purpose	Conclusion supported
Random sampling	Select a representative sample from a defined population	Generalization to the sampled population
Random assignment	Assign experimental units to treatments through chance	Stronger evidence about treatment effects

A study may use one, both or neither form of randomization. The scope of the conclusion should match the design that was used.

Independence of Observations

Many standard errors and probability models assume that the observations are independent.

- In a random sample, one selected individual should not determine another individual's response
- Repeated measurements from the same individual are usually dependent
- Animals in the same litter, patients in the same clinic or measurements from nearby sites may be correlated

When sampling without replacement from a finite population, the condition

$$n \leq 0.10N$$

often supports treating observations as approximately independent. Dependence may require a different analysis.

Practical Problems After Data Collection Begins

Even a well-planned study can encounter difficulties.

- Nonresponse can make respondents systematically different from nonrespondents
- Attrition can create bias when participants who leave differ from those who remain
- Measurement problems can introduce systematic error outside the sampling model
- Clustering or unequal selection probabilities can invalidate a simple random-sample formula

A probability calculation cannot correct a fundamental mismatch between the study design and the statistical model.

Choosing a Population Model

The response variable determines the starting probability model.

Response	**Possible model**	**Parameter**
Continuous measurement	Normal model or a model justified through a sampling distribution	Population mean μ
Binary outcome	Binomial model	Population probability p
Event count over exposure	Poisson model	Population rate λ

Model assumptions concern more than the shape of a graph. They also describe independence, a stable probability or rate and the relevant population.

Sample Size and the Normal Approximation

The amount of non-Normality that a procedure can tolerate depends partly on the sample size.

- With a small sample, strong skewness or outliers can make Normal-based inference unreliable
- With a moderate sample, a roughly symmetric population may be sufficient
- With a large sample, the Central Limit Theorem supports an approximately Normal sampling distribution under broad conditions

There is no universal sample-size cutoff. Stronger skewness and heavier tails generally require more observations.

Outliers and Nonresistant Statistics

The sample mean $\bar{x}$ and sample standard deviation s are sensitive to extreme observations.

An outlier can change the estimate, increase the standard error and alter the conclusion of a test.

Before performing inference:

- Check for data-entry and measurement errors
- Examine a suitable graph of the observations
- Investigate scientifically plausible reasons for unusual values
- Consider another method when a small sample contains severe skewness or influential outliers

An observation should not be removed solely because its removal changes the conclusion.

A Conditions Checklist

Before reporting an inferential result, consider the following questions.

1. What population or treatment comparison does the study address?
2. How were individuals selected and, when applicable, assigned to treatments?
3. Are the observations sufficiently independent?
4. Does the probability model match the response and the study process?
5. Are outliers, missing observations or measurement problems present?
6. Does the analysis account for the actual sampling or experimental design?

These questions determine whether the standard error, confidence interval and P-value have a defensible interpretation.

The General Form of a Confidence Interval

Many confidence intervals have the form

$$\text{estimate} \pm \text{margin of error}$$

The margin of error combines a critical value and a standard error.

$$\text{margin of error} = \text{critical value} \times \text{standard error}$$

For a one-sample z confidence interval for a population mean,

$$m = z^* \frac{\sigma}{\sqrt{n}}$$

The estimate determines the centre. The margin of error determines the half-width.

From a Sampling Distribution to an Interval

Suppose the standardized statistic follows a Standard Normal distribution:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

For a 95% procedure,

$$P\left(-1.96 \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq 1.96\right) = 0.95$$

Rearranging the inequalities gives

$$P\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

This probability statement describes the interval-producing procedure before the sample is observed.

Normal Critical Values

The critical value z^* leaves the chosen confidence level in the centre of the Standard Normal distribution.

Confidence	**Central area**	**Area in each tail**	z^*
90%	0.90	0.05	1.645
95%	0.95	0.025	1.960
99%	0.99	0.005	2.576

This is the same area reasoning used for Normal probabilities in Chapter 11. Higher confidence moves the cutoffs farther into the tails.

Long-Run Coverage

The population parameter remains fixed. Different random samples produce different estimates and therefore different intervals.

- A 95% confidence procedure captures the true parameter in approximately 95% of repeated random samples under the model
- Approximately 5% of the intervals miss the parameter
- After one interval has been calculated, the endpoints are fixed

Interpretation

Confidence describes the long-run performance of the method. It does not assign a probability to the fixed parameter after the data have been observed.

Confidence and Precision

Two desirable features of an interval compete with one another.

- A high confidence level gives a procedure with a high long-run coverage rate
- A small margin of error gives a more precise estimate of the parameter

With the same data, higher confidence requires a larger z^* and a wider interval.

Trade-Off

Reducing the confidence level can narrow the interval, but the procedure will capture the true parameter less often over repeated samples.

Population Variability and Precision

The margin of error increases with the population standard deviation.

$$m = z^* \frac{\sigma}{\sqrt{n}}$$

- A smaller σ produces less sample-to-sample variation in $\bar{x}$
- A larger σ makes the population mean more difficult to estimate precisely

Natural biological variation usually cannot be removed. Measurement variation can sometimes be reduced through better instruments, procedures or training.

Sample Size and the Square-Root Relationship

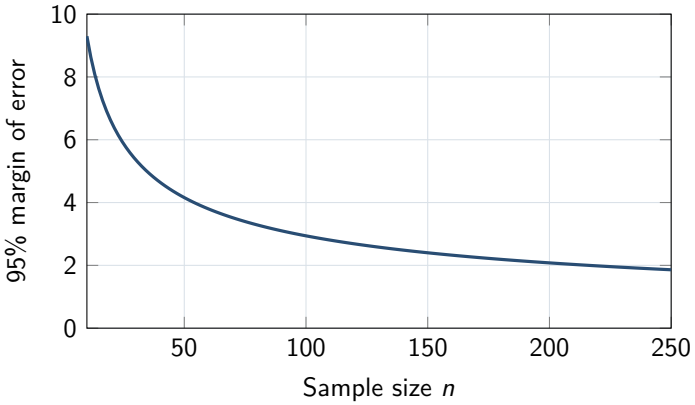
For a fixed confidence level and population standard deviation,

$$m \propto \frac{1}{\sqrt{n}}$$

Change in sample size	New standard error	Effect on margin
$n \to 4n$	One-half as large	Margin is halved
$n \to 9n$	One-third as large	Margin is divided by 3
$n \to 100n$	One-tenth as large	Margin is divided by 10

Doubling the sample size multiplies the margin by $1/\sqrt{2} \approx 0.707$ rather than cutting it in half.

Margin of Error and Sample Size



For this illustration,

$$\sigma = 15$$

The largest gains in precision occur when the sample is initially small.

Later increases still help, but each additional observation produces a smaller reduction in the margin.

What the Margin of Error Measures

The margin of error describes uncertainty from random sampling under the assumed model.

It reflects how far the sample statistic may fall from the population parameter because another random sample would give a different statistic.

Covered by the Margin of Error

Random sample-to-sample variation generated by the stated sampling process

A correctly calculated margin of error can accompany a misleading conclusion when the observations are biased or the model does not represent the study.

Errors Outside the Margin of Error

The usual margin of error does not measure uncertainty caused by

- Undercoverage, voluntary response or nonresponse
- Systematic attrition during a study
- Leading questions or inaccurate self-reports
- Poorly calibrated instruments or inconsistent measurements
- Confounding in an observational study
- A model that does not match the design or population

These sources of error can exceed the reported sampling error. A narrow interval does not compensate for biased data.

A P-Value as a Model-Based Probability

A hypothesis test temporarily assumes that the null hypothesis and model conditions are true.

$$P\text{-value} = P(\text{statistic at least as extreme as observed} \mid H_0)$$

- A small value means that the observed result lies in an unusual part of the null distribution
- A large value means that the observed result remains reasonably compatible with the null model

The calculation evaluates the data under H_0. It does not calculate the probability that H_0 is true.

The Test Statistic Locates the Data

For a one-sample z test of a population mean,

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

- The numerator is the observed difference from the null value
- The denominator is the standard error expected under repeated sampling
- The resulting z value measures the difference in standard-error units

The P-value is the relevant tail area beyond the observed test statistic under the null distribution.

Evidence Forms a Continuum

Smaller P-values indicate greater incompatibility between the observed statistic and the null model.

A significance level such as $\alpha = 0.05$ provides a decision rule, but it does not create a sharp scientific distinction between nearby results.

Example

$P = 0.048$ and $P = 0.052$ represent nearly the same strength of evidence, although a fixed 5% rule gives different formal decisions.

The exact P-value, effect estimate, confidence interval and consequences of a decision should inform the conclusion.

The Alternative Hypothesis Determines the Tail

The direction of the scientific question determines which probabilities count as at least as extreme.

- $H_a : \mu > \mu_0$ uses the upper tail
- $H_a : \mu < \mu_0$ uses the lower tail
- $H_a : \mu \neq \mu_0$ combines two disjoint tails

For a symmetric null distribution,

$$P_{\text{two-sided}} = 2P(Z \geq |z_{\text{obs}}|)$$

Choosing the direction after seeing the data understates uncertainty and does not preserve the intended error rate.

Type I and Type II Errors

	H_0 **is true**	**A specified alternative is true**
Fail to reject H_0	Correct decision	Type II error
Reject H_0	Type I error	Correct decision

$$P(\text{Type I error}) = \alpha \qquad P(\text{Type II error}) = \beta$$

The relative seriousness of the two errors depends on their scientific, medical or policy consequences.

Statistical Significance and Sample Size

For a one-sample z test,

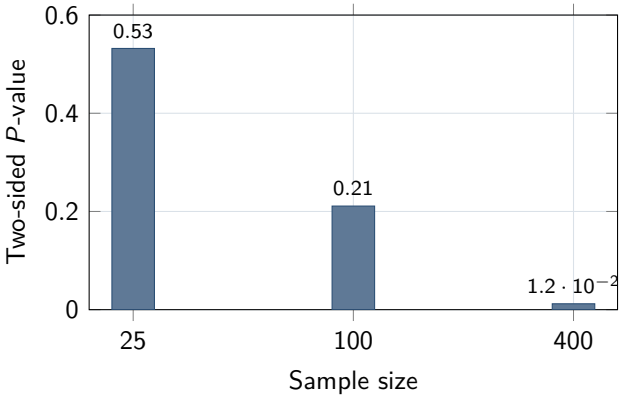
$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{\text{observed effect}}{\text{chance variation}}$$

The test statistic depends on the size of the observed effect, the population variability and the sample size.

A very large sample can identify a small effect as statistically significant. A small or highly variable sample may fail to detect an effect that matters in practice.

The Same Effect at Different Sample Sizes

Suppose $\bar{x} - \mu_0 = 0.30$ and $\sigma = 2.4$ in each study.



The estimated effect stays fixed.

As n increases, the standard error decreases and the same effect becomes less compatible with H_0.

The practical size of the effect has not changed.

Statistical Significance and Practical Importance

Question	**Useful information**
Are the data difficult to explain under H_0?	P-value and model conditions
How large might the effect be?	Effect estimate and confidence interval
Would the effect matter in practice?	Scientific thresholds, consequences, costs and benefits

A complete analysis reports the size and uncertainty of the effect rather than only whether a significance threshold was crossed.

What a P-Value Does Not Mean

A P-value is calculated under the null hypothesis and the model assumptions.

It is not

- The probability that H_0 is true
- The probability that the result occurred only because of chance
- The size or importance of the observed effect
- A substitute for checking the study design and model conditions

A large P-value indicates compatibility with the null model. It does not establish that the null hypothesis is correct.

Multiple Tests and the Complement Rule

Suppose m independent tests use significance level α and all null hypotheses are true.

$$P(\text{no false positives}) = (1 - \alpha)^m$$

The complement rule gives

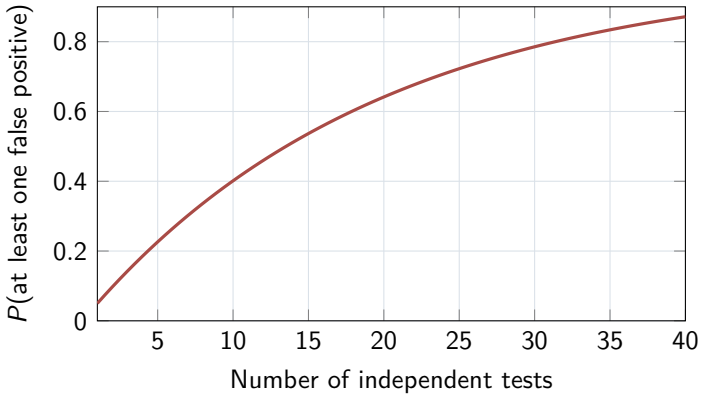
$$P(\text{at least one false positive}) = 1 - (1 - \alpha)^m$$

For $m = 20$ and $\alpha = 0.05$,

$$1 - (0.95)^{20} \approx 0.642$$

This calculation explains why selective reporting can produce apparently significant findings even when every null hypothesis is true.

False Positives Across Multiple Tests



Each test still has Type I error probability 0.05.

The family-level chance of at least one false positive increases as more questions are tested.

The calculation assumes independent tests.

Inference Within the Scientific Process

Statistical inference supports a transparent scientific process.

1. State the research question and analysis plan before examining the results
2. Use a design that connects the data to the intended population or causal question
3. Examine the observations and assess the model conditions
4. Report estimates, uncertainty and exact P-values in context
5. Replicate important findings with new data when possible

Replication reduces the chance that a conclusion depends on an unusual sample, an unnoticed design problem or flexible analysis choices.

Planning a Confidence Interval

For a one-sample z confidence interval,

$$m = z^* \frac{\sigma}{\sqrt{n}}$$

Solving for the required sample size gives

$$n = \left(\frac{z^* \sigma}{m} \right)^2$$

The value must be rounded up so that the achieved margin of error does not exceed the target.

A plausible value of σ can come from previous research or a pilot study.

Population Size and Required Sample Size

The usual sample-size formula depends on

- The confidence level through z^*
- The expected population variability through σ
- The desired precision through m

Population size does not appear when the population is much larger than the sample. The same planned sample size can provide the same margin of error in two large populations when their sampling designs and variability are comparable.

A finite-population correction may be used when the sample forms a substantial fraction of the population.

Planning a Hypothesis Test

Planning a test requires decisions about

- **Significance level α:** the tolerated Type I error probability
- **Meaningful effect:** the smallest departure from H_0 that matters in context
- **Power:** the desired probability that the test detects that effect

These choices should be made before data collection. They connect the statistical design to the consequences of missing an important effect or incorrectly declaring one.

Power and Type II Error

The power of a test against a specified alternative is

$$\text{Power} = P(\text{reject } H_0 \mid \text{specified alternative is true})$$

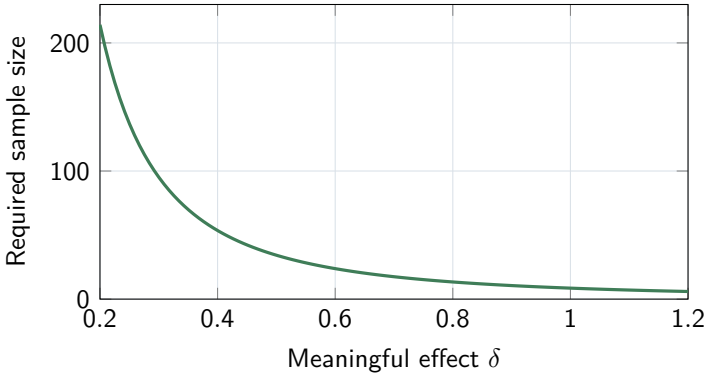
If β is the Type II error probability for that alternative, then

$$\text{Power} = 1 - \beta$$

Power generally increases with a larger sample, a larger effect, lower population variability or a larger significance level. A two-sided test usually needs more observations than a one-sided test with the same target power.

Effect Size and Required Sample Size

For this illustration, $\alpha = 0.05$, power is 0.90, $\sigma = 1$ and the alternative is one-sided.



Smaller effects are more difficult to distinguish from sampling variation.

A study therefore needs more observations when it aims to detect a smaller meaningful effect.

Applications Across Probability Models

The next examples use the same inferential logic with different probability models.

Outcome	**Model-based comparison**	**Practical check**
Binary treatment response	Compare an observed success count with a binomial null model	Independence and a common response probability
Disease-event count	Compare an observed count with a Poisson null model	Stable rate, exposure and absence of clustering
Continuous measurement	Compare a sample mean with a Normal sampling model	Study design, distributional shape and variability

The calculation changes with the data type. The need to evaluate assumptions and interpret the result in context does not change.

Example 15.1: The Scope of Generalization

Two researchers recruit students from introductory courses at one university.

Neurobiology Study

A researcher studies responses to an optical illusion. Basic visual mechanisms may be similar across many groups, so subject-matter knowledge may support a cautiously broader interpretation.

Attitude Survey

A researcher studies opinions about human-participant research. Age, education, program and campus culture may affect responses, so one class does not represent adults or university students generally.

Neither class is a random sample. The defensible scope differs because the research questions differ.

Example 15.2: Binomial Model and Relevant Tail

A veterinary study evaluates whether a treatment reaches a proposed response probability of 0.80. Among 60 independently treated dogs, 41 show the response defined in the study protocol.

$$H_0 : p = 0.80 \qquad H_a : p < 0.80$$

Let X be the number of dogs that respond. Under the null hypothesis,

$$X \sim \text{Bin}(60, 0.80)$$

The lower-tail alternative makes counts of 41 or fewer at least as extreme as the observed result. Therefore, the P-value is

$$P(X \leq 41 \mid H_0) = P(X \leq 41 \mid p = 0.80)$$

The vertical bar means that the probability is calculated under the null model.

Example 15.2: Exact Binomial Calculation

For a binomial random variable,

$$P(X = x \mid H_0) = \binom{60}{x}(0.80)^x(0.20)^{60-x}$$

The lower-tail probability adds the probabilities for all counts from 0 through 41:

$$\begin{aligned} P(X \leq 41 \mid H_0) &= \sum_{x=0}^{41} \binom{60}{x}(0.80)^x(0.20)^{60-x} \\ &\approx 0.022068 \\ &\approx 0.0221 \end{aligned}$$

If $p = 0.80$, a response count of 41 or fewer would occur in approximately 2.21% of comparable studies.

Example 15.2: Exact Binomial Calculation in R

Lower-tail probability

```
# Null model: X ~ Binomial(60, 0.80)
n_dogs <- 60
p_null <- 0.80
x_observed <- 41

p_value <- pbinom(
  q = x_observed,
  size = n_dogs,
  prob = p_null
)

p_value
# [1] 0.02206845
```

The argument `q = 41` includes every count from 0 through 41. The result agrees with the exact summation.

Example 15.3: Poisson Model and Relevant Tail

A wildlife-health program historically records an average of 12 confirmed cases of a disease per month. During one month, the program records 20 cases.

$$H_0 : \lambda = 12 \qquad H_a : \lambda > 12$$

Let X be the number of confirmed cases in one month. Under the stable-rate null model,

$$X \sim \text{Pois}(12)$$

The upper-tail alternative makes counts of 20 or more at least as extreme as the observed count. The P-value is

$$P(X \geq 20 \mid H_0) = P(X \geq 20 \mid \lambda = 12)$$

Example 15.3: Exact Poisson Calculation

For a Poisson random variable,

$$P(X = x \mid H_0) = e^{-12}\frac{12^x}{x!}$$

It is shorter to use the complement of the cumulative probability through 19:

$$\begin{aligned} P(X \geq 20 \mid H_0) &= 1 - P(X \leq 19 \mid H_0) \\ &= 1 - \sum_{x=0}^{19} e^{-12}\frac{12^x}{x!} \\ &\approx 0.021280 \\ &\approx 0.0213 \end{aligned}$$

If the mean remained 12 cases per month, a count of 20 or more would occur in approximately 2.13% of comparable months.

Example 15.3: Exact Poisson Calculation in R

Upper-tail probability

```
# Null model: X ~ Poisson(12)
lambda_null <- 12
x_observed <- 20

p_value <- ppois(
  q = x_observed - 1,
  lambda = lambda_null,
  lower.tail = FALSE
)

p_value
# [1] 0.02127977
```

With `lower.tail = FALSE`, R calculates $P(X > 19)$, which is the same event as $P(X \geq 20)$ for an integer-valued count.

Example 15.4: Confidence Level and Margin of Error

For a study with

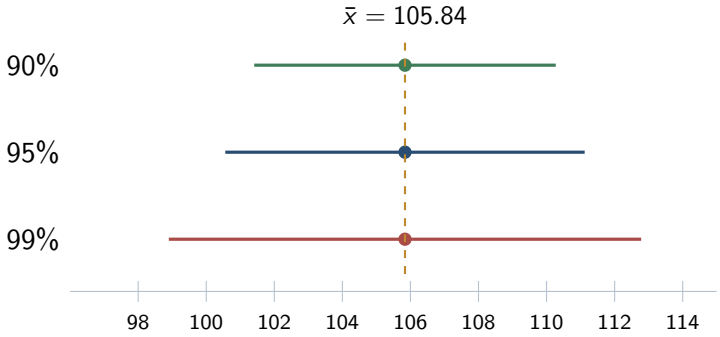
$$\bar{x} = 105.84, \qquad \sigma = 15, \qquad n = 31$$

the confidence level changes the critical value and the interval width.

Confidence	z^*	**Margin**	**Interval**
90%	1.645	4.43	(101.41, 110.27)
95%	1.960	5.28	(100.56, 111.12)
99%	2.576	6.94	(98.90, 112.78)

All three intervals share the same centre because they use the same sample mean.

Comparing Confidence Levels



Higher confidence leaves more probability in the central Normal area, so the interval must extend farther from the estimate.

Example 15.4: Increasing the Sample Size

At 95% confidence, the margin for $n = 31$ is

$$m_{31} = 1.960\frac{15}{\sqrt{31}} \approx 5.28$$

If the sample size increases to 62 while σ remains 15, then

$$m_{62} = 1.960\frac{15}{\sqrt{62}} \approx 3.73$$

$$\frac{m_{62}}{m_{31}} = \sqrt{\frac{31}{62}} = \frac{1}{\sqrt{2}} \approx 0.707$$

Doubling the sample size reduces the margin by about 29.3%. A new sample would usually also change the sample mean and therefore the interval centre.

Example 15.5: A Large Social-Media Experiment

A randomized experiment altered the emotional content shown in the News Feeds of 689,003 users for one week.

The reported tests produced very small P-values:

$$z = -4.78 \qquad \text{and} \qquad z = -7.219 \qquad P < 0.001$$

The estimated change was approximately one fewer emotional word per 1000 words written during the following week.

Interpretation

The large sample made the analysis sensitive to a very small average change. The P-values provide evidence against a zero-effect model but do not establish a large effect.

Example 15.5: Practical Importance

The social-media experiment raises questions that a P-value cannot answer.

- Does using fewer emotional words represent a meaningful change in emotion?
- Could users repeat language that recently appeared in their feeds?
- Does the experimental change resemble ordinary use of the platform?
- Could a small individual effect matter when applied across a large population?

Evaluating importance requires the effect estimate, its uncertainty, the quality of the measurement and knowledge of the setting.

Example 15.6: Planning a Body-Temperature Study

Researchers want to estimate the mean body temperature of healthy adults within 0.05°F at 95% confidence. Previous research suggests

$$\sigma = 0.6^\circ\mathrm{F} \qquad \text{and} \qquad z^* = 1.960$$

The required sample size is

$$n = \left(\frac{1.960(0.6)}{0.05}\right)^2 = 553.1904$$

The study must recruit at least

$$\boxed{n = 554}$$

Rounding down would produce a margin of error slightly larger than the target.

Example 15.7: Power in a Sweetness Study

A quality-control study tests

$$H_0 : \mu = 0 \qquad H_a : \mu > 0$$

with $\sigma = 1$, $\alpha = 0.05$, $n = 10$ and a meaningful mean loss of $\mu = 0.8$ point.

The critical sample mean is

$$\bar{x}_{\text{crit}} = 1.645 \left(\frac{1}{\sqrt{10}} \right) \approx 0.520$$

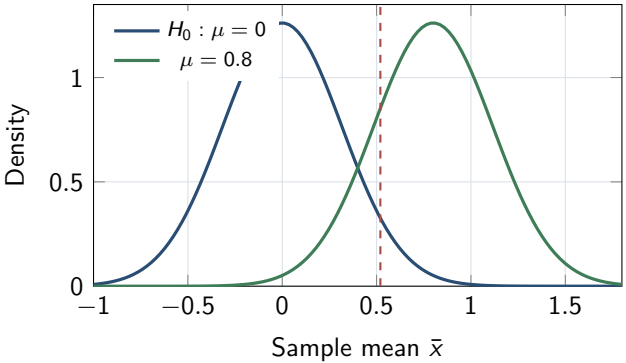
The power at $\mu = 0.8$ is

$$z = \frac{0.520 - 0.8}{1/\sqrt{10}} \approx -0.886$$

$$\text{Power} = P(Z \geq -0.886) \approx 0.812$$

The initial design does not reach the target power of 0.90.

Example 15.7: Error Probabilities



The rejection region begins at $\bar{x} = 0.520$.

$$\alpha = 0.05$$

$$\beta = 1 - 0.812 = 0.188$$

$$\text{Power} = 0.812$$

The two probabilities come from different sampling distributions.

A Consistent Inferential Workflow

1. Identify the population, parameter and scientific question
2. Describe how the observations were selected or treatments were assigned
3. Match the response to a probability model and check its assumptions
4. Choose the statistic and determine its sampling distribution
5. Calculate an interval or a test statistic and P-value
6. Report the effect size, uncertainty, limitations and practical meaning

Probability supports the calculation. Study design and subject-matter reasoning determine the scope of the conclusion.

Chapter 15 Summary

Probability provides the mathematical foundation for inference.

- Probability rules determine tail areas, error rates and multiple-testing risks
- Normal, binomial and Poisson models connect different data types to inferential questions
- The study design and model conditions determine whether an inferential calculation is trustworthy
- Confidence intervals describe effect size and sampling uncertainty
- Statistical significance depends on the effect, variability and sample size
- Study planning connects precision, meaningful effects, error probabilities and power

A clear conclusion reports the estimate, uncertainty, study limitations and practical meaning together.